5. Project Development Phase

Script-Controlled ACL: Restrict Record Access Based on Field Value

Milestone 1: Creation of Users and Roles

User creation

1. Log in to the ServiceNow instance with admin access.
2. Navigate to User Administration > Users and click **New**.
3. Enter User ID: **EEE User**, First name: **EEE**, Last name: **User**, Email: **eeeuser@gmail.com**.
4. Save the record.

Role creation

1. Navigate to User Administration > Roles and click **New**.
2. Name: **bb1**; add a brief description; click **Submit**.
3. Repeat for **bb2, bb3 and bb4**.
4. Assign all four roles to the EEE User.

Table Creation

1. In the Application Navigator open System Definition > Tables and click **New**.
2. Label: **Institution Details**; Name: **u_institution_details**; Extends: false. Click **Submit**.
3. Add fields: Student Roll Number (Auto Number), Student Name (Reference: User), Faculty Name (Reference: User), Branch (Choice: ECE, EEE, CSE), Email (String), Phone Number (String), Description (Multi String).
4. Save the table and create several records with different Branch values (ECE, EEE, CSE).

READ ACL

1. Open System Security > Access Control (ACL) and elevate the **security_admin** role.
2. Click **New**.
3. Type: **record**; Operation: **read**; Name: **u_institution_details**; Active: **true**; Advanced: **true**.
4. In the Requires role related list add **bb1**.
5. Add the data condition **Branch is EEE**.
6. Paste the script below into the Script field and click **Submit**.

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }
    // Allow only EEE branch users to see EEE records
    if (gs.hasRole('bb1')) {
        return true;
    }
    // Deny access for all others
    return false;
})();
```

CREATE, WRITE and DELETE ACLs

For each ACL click **New**, set Type to record, Name to u_institution_details, Active to true, add the required role in the Requires role related list, add no data condition, and save.

ACL	Operation	Required role
Create ACL	create	bb2
Write ACL	write	bb3
Delete ACL	delete	bb4

Access Testing Setup

- Impersonate the EEE User and open the list by typing **u_institution_details.list** in the All menu filter.
- Impersonate the admin and open the same list to compare results.